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Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No: 6
Title: Creating HR analytics dashboards

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Creating HR analytics dashboards

Github Link: <https://github.com/siddiquinadeem51/Data-Analytics-For-Business-Projects>

Problem Statement :

You are an HR Data Analyst in a multinational organization. The HR department wants insights into employee demographics, hiring trends, attrition, and workforce distribution. Develop an HR analytics dashboard to support workforce planning and employee management.

Methodology :

Data Visualization (Creating Dashboard With Changing Values):

1. First I imported IBM HR analytics dataset into Power BI
2. Then performed data cleaning and transformation
3. After that I created calculated measures using DAX
4. The designed an interactive dashboard using:
 - KPI Cards
 - Bar Charts
 - Column Charts
 - Pie Charts
 - Slicers (Filters)

DAX Measures Used :

1. Total Employee:

- Total Employee = COUNT('IBM HR'[EmployeeNumber])

Screenshot Image:

```
1 Total Employee = COUNT('IBM HR'[EmployeeNumber])
```

2. Total Attrition:

- Total Attrition = CALCULATE(COUNT('IBM HR'[EmployeeNumber]), 'IBM HR'[Attrition] = "Yes")

Screenshot Image:

```
1 Total Attriton = CALCULATE(COUNT('IBM HR'[EmployeeNumber]), 'IBM HR'[Attrition]="Yes")
```

3. Attrition Rate:

- Attrition Rate = DIVIDE([Total Attrition], [Total Employee], 0)

Screenshot Image:

```
1 Attrition Rate = DIVIDE([Total Attriton],[Total Employee],0)
```

4. Average Monthly Rate:

- Average Monthly Rate = AVERAGE('IBM HR'[DailyRate])

Screenshot Image:

```
1 Average Monthly Rate = AVERAGE('IBM HR'[DailyRate])
```

Dashboard Images :

Image 1st:

This image shows the complete HR Analytics dashboard without any filters applied.

- Displays overall KPIs:
- Total Employees: 1.47K
- Total Attrition: 237
- Attrition Rate: 16.12%
- Shows attrition distribution by:
- Job Role
- Department
- Education Field
- Includes Gender Distribution and Employee Tenure Distribution
- Provides a high-level overview of workforce and attrition patterns

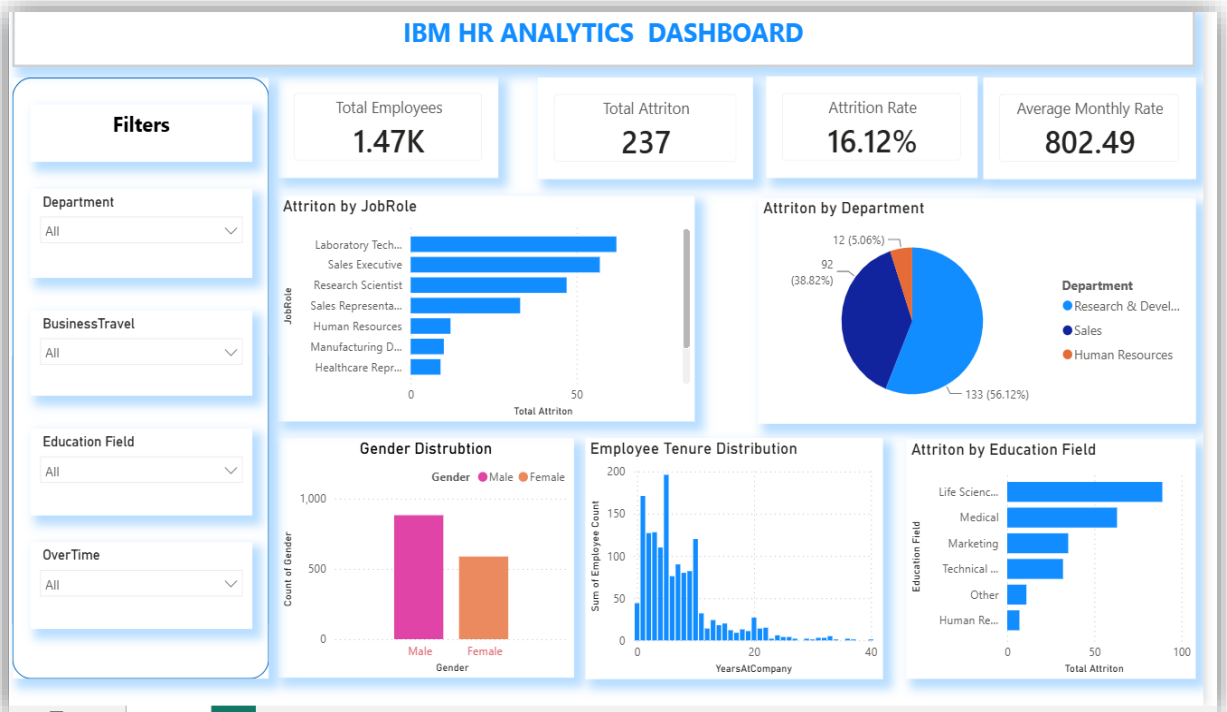


Image 2nd:

This image shows the dashboard after applying some filters (subset of employees).

- KPIs dynamically change:
- Total Employees reduced to 63
- Attrition Rate increased to 19.05%
- Visuals update automatically based on selected data
- Helps analyze specific group behavior
- Demonstrates interactive filtering capability

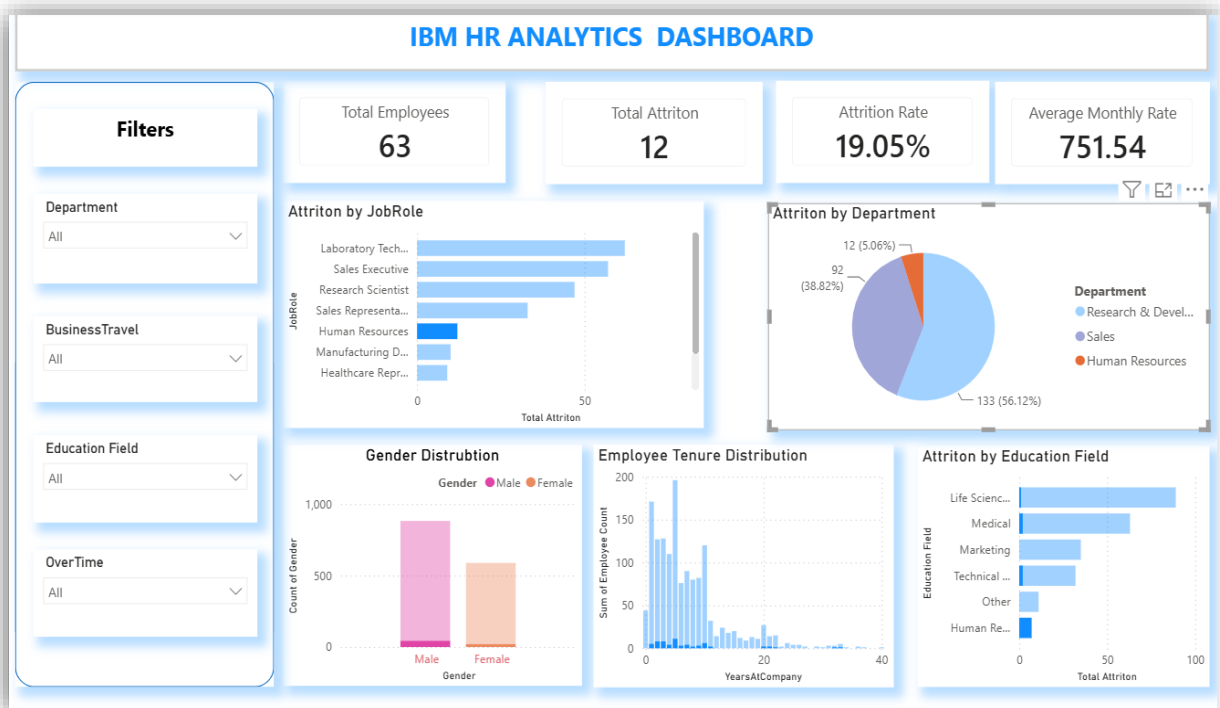
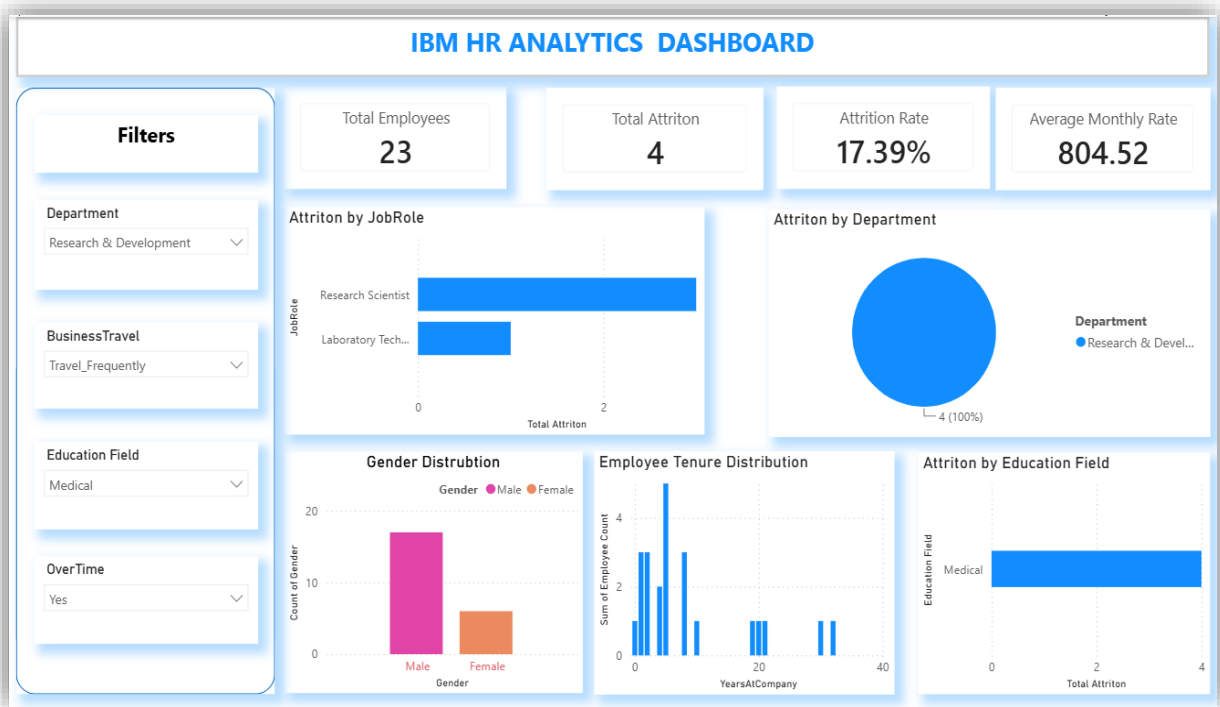


Image 3rd:

This image shows the dashboard with multiple filters applied:

- Department: Research & Development
- Business Travel: Travel Frequently
- Education Field: Medical
- Over Time: Yes
- KPIs further change:
 - 1.Total Employees: 23
 - 2.Attrition Rate: 17.39%
- Charts now reflect a very specific employee segment
- Enables deep-dive analysis of targeted workforce groups



Business Insights:

- The attrition rate of the organization is quite high (~16%) that denotes difficulties in retaining employees.
- There is attrition in certain positions within the company.
- Most of the employees who tend to leave the company have low tenure (initial years).
- With filtering, it becomes clear that there is attrition among various categories of employees.

Conclusion:

The HR Analytics dashboard effectively visualizes employee data and attrition patterns using interactive Power BI visuals. It highlights key issues such as high attrition in specific roles, departments, and early employee tenure. The use of filters allows detailed analysis of different workforce segments. Overall, the dashboard supports data-driven decision-making and helps HR teams improve employee retention and workforce planning.